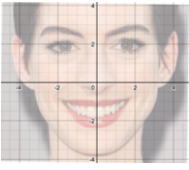


Seeing Symmetry: Faces and Functions



Symmetry in everyday language refers to a sense of harmonious and beautiful proportion and balance. There's even research that says that we rate symmetric faces as being more attractive! Today we will investigate further what it means for functions to be symmetric and how we might predict the symmetry of a graph from its equation.

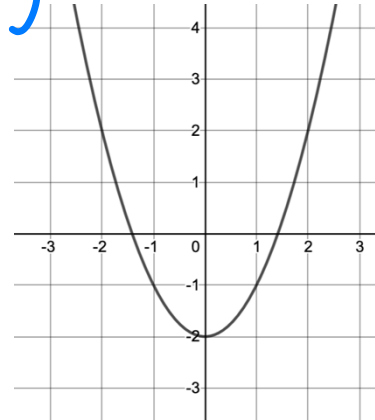
$f(x)$ is an even function.

1. The graph of $f(x) = x^2 - 2$ is below. What do you notice about the symmetry of $f(x)$?

It has y-axis symmetry

2. Fill in the table of values for $f(x)$.

x	$f(x)$
-4	14
-2	2
0	-2
2	2
4	14



3. What patterns do you notice in the ordered pairs?

Both 4 and -4 output the same y-value.
As do -2 and 2. Opposite x-values give same y-value.

4. Given an equation of a function, how could you check if the function has y-axis symmetry without looking at a graph?

See if plugging in opposite x-values gives the same output.

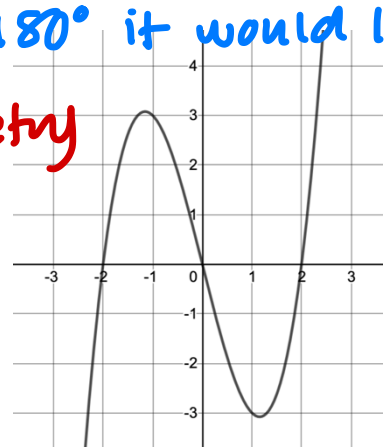
5. The graph of $g(x) = x^3 - 4x$ is below. What do you notice about the symmetry of $g(x)$?

$g(x)$ is an odd function.

6. Fill in the table of values.

x	$g(x)$
-4	-48
-1	3
0	0
1	-3
4	48

origin symmetry



7. What patterns do you notice in the ordered pairs?

4 and -4 give opposite values.
So do 1 and -1.

8. Given an equation of a function, how could you check if the function is symmetric across the origin without looking at a graph?

Plug in opposite x-values and see if you get opposite y-values

9. How symmetric is Anne Hathaway's face? (See photo above). How can you tell?

Very symmetric.
(2,2) and (-2,2) both line up w/ the corner of her eye.

Lesson 1.6 – Even and Odd Functions

QuickNotes

A function is even if it has y-axis symmetry.

$f(-x) = f(x)$

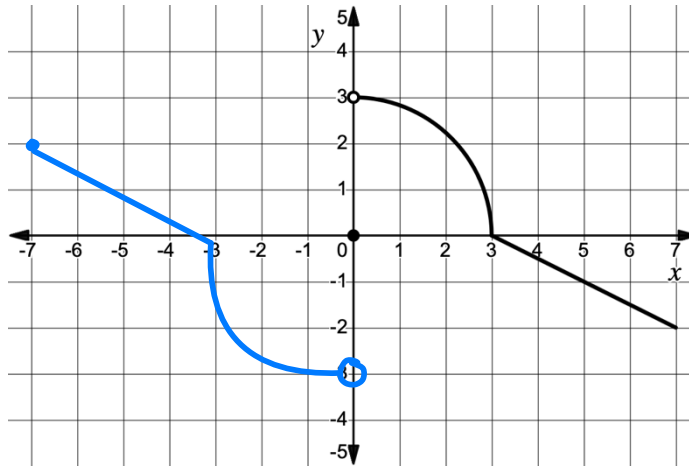
A function is odd if it has origin symmetry (180° rotational symmetry).

$f(-x) = -f(x)$

To prove symmetry algebraically:
 Plug in $(-x)$ for x and simplify expression.
 Compare to original.

Check Your Understanding

1. Half of the graph of an odd function is shown below. Sketch the other half of the function.



2. If $(3, -18)$ is on the graph of an even function, what other point is guaranteed to be on the graph?

$$(-3, -18)$$

3. Is $y = x^5 + x^3 + 3$ an even function, an odd function, or neither? Prove your answer algebraically.

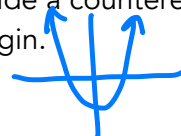
$$(-x)^5 + (-x)^3 + 3 = -x^5 - x^3 + 3$$

This is neither $x^5 + x^3 + 3$ nor $-(x^5 + x^3 + 3)$.
 Thus y is neither even nor odd.

4. True or false: If the statement is true, explain why it must be true using the definitions of even and odd functions. If it is false, provide a counterexample.

- a. All even functions go through the origin.

False



$$f(x) = x^2 - 2$$

- b. If each term in a polynomial function has an even exponent, the function is even.

True; raising $(-x)$ to an even exponent makes it positive so $f(-x)$ would be equivalent to $f(x)$.